

Gibbs-Anchored Density-Ratio Validation for the Three-Mode NLS Stationary State

14 September 2026

Abstract

We tested a preregistered Gibbs-anchored representation of the five-dimensional stationary density, using independent-trajectory train, validation and blind test splits. The exact equilibrium known-answer stage passed, but neither of the two equilibrium-admissible model classes passed the driven validation gates. The driven blind split therefore remained sealed. The result is a methodological partial result, not a numerical solution of the NESS density.

1 Frozen question and method

The previous free normalized neural density did not reproduce the known Gibbs stationary equation. Here the equilibrium density is built in as a reference,

$$\rho_\eta(X) = \rho_0(X) \frac{\exp f_\eta(X)}{Z_\eta}, \quad \rho_0(X) \propto \exp[-E(X)/5]. \quad (1)$$

The network therefore learns only the density ratio. Inputs use bounded action features $I_j/(I_j + s_j)$ and periodic Fourier features of the two bond angles. The preregistered candidates were a linear model and two tanh MLPs with widths 32×32 and 64×64 , each with seeds 5201–5203. All inputs are the existing projection-free float64 trajectories; no new stochastic simulation was performed.

Streams were split at trajectory level into 32 training, 16 validation and 16 blind-test trajectories. For the equilibrium known-answer test, each split was further divided deterministically into disjoint pseudo-target and reference streams. The exact normalized log density ratio is then zero.

2 Equilibrium known-answer gate

The linear and 64×64 candidates passed every validation gate for all three seeds. The 32×32 family was rejected because one seed had median $|\mathcal{L}^\dagger \rho / \rho| = 0.1227$, above the frozen 0.10 bound. After freezing the admissible set, the equilibrium blind split was opened once. All six retained fits passed. Blind centered log-ratio RMS was 0–0.00893, ROC AUC was 0.5000–0.5026, and reference-weight ESS fraction was 0.99992–1.00000.

This establishes that the Gibbs anchor and density-ratio implementation retain the exact equilibrium law without manufacturing a spurious correction. It does not by itself establish that the correction family is expressive enough for the driven state.

Table 1: Fine-timestep driven validation. The frozen residual gates were 0.10 for the median and 0.50 for the 90th percentile.

Model	seed	ESS frac.	FP median	FP p90	marginals	moments	overall
linear	5201	0.9613	0.8656	3.9998	pass	fail	fail
linear	5202	0.9608	0.8702	4.0111	pass	fail	fail
linear	5203	0.9598	0.8535	3.9549	pass	fail	fail
MLP64	5201	0.9550	0.9824	4.4574	pass	fail	fail
MLP64	5202	0.9542	0.8930	4.0756	pass	fail	fail
MLP64	5203	0.9617	0.8849	4.0822	pass	fail	fail

3 Driven validation

Importance support was not the limitation: every fit retained more than 95% of the nominal reference-sample ESS. All frozen marginal-TV gates also passed; the largest TV distance, for the joint two-angle histogram, was 0.0377–0.0420 against the 0.05 bound. Across-seed log-ratio disagreement also passed.

The failure was instead concentrated in the stationary equation and current-sensitive observables. The median pointwise residual was 0.853–0.982, roughly nine times the gate, and its 90th percentile was 3.95–4.46. Table 2 shows the two reproducible moment failures.

Table 2: Current-sensitive validation moments. The z values use whole-stream bootstrap standard errors; the family-wise threshold was 3.748.

Model/seed	target	$I_1 I_2 \sin \theta_1$			$I_2 I_3 \sin \theta_3$		
		ratio	z		ratio	z	
linear/5201	0.23046	0.10122	-29.59		-0.22980	-0.10395	26.17
linear/5202	0.23046	0.10274	-26.88		-0.22980	-0.09667	24.97
linear/5203	0.23046	0.09978	-29.39		-0.22980	-0.09377	26.68
MLP64/5201	0.23046	0.15946	-14.55		-0.22980	-0.12449	20.33
MLP64/5202	0.23046	0.12236	-23.47		-0.22980	-0.10582	24.27
MLP64/5203	0.23046	0.12775	-22.64		-0.22980	-0.09571	27.05

4 Verdict and claim boundary

No model passed the driven validation gate, so the frozen selector recorded that there was no admissible driven candidate. In accordance with the protocol, the driven blind split was not read and the coarse-timestep stage was not run.

The result is therefore **PARTIAL**: the Gibbs anchor fixes the previous equilibrium support/normalization problem, but the tested density-ratio fits do not solve the driven stationary Fokker–Planck equation. Agreement of coarse marginals cannot be promoted to a joint NESS-density claim, because the same fits miss the stationary current observables by 15–30 standard errors.

The next density protocol should change the optimization and validation target so that held-out stationarity and current-sensitive weak constraints influence model selection. The present thresholds must not be relaxed post hoc.

Reproducibility. All 256 trajectories match the frozen byte counts and SHA-256 manifest. Exact commands, seeds, source hashes, raw metrics and the I/O amendment accompany this report.